

Seed robustness of the gate-unfreeze QFT

100 random-initialisation seeds $\times$ three unfreeze orderings, top-20% training objective, DIV2K-8q

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Question

Does the gate-unfreeze QFT endpoint depend on the random draw? We retrain the **QFTBasis(8,8)** operator from **Haar-random** init under the block-growth (**bg**), left $\rightarrow$ right (**lr**) and right $\rightarrow$ left (**rl**) unfreeze orderings, for 100 seeds each. Every seed s reseeds **everything trainable** — the Haar gate init, the 50-image training-batch subsample (from the fixed 500-image pool), and the training RNG — while the held-out **test set is held fixed** (canonical seed 42) so the endpoint-PSNR spread reflects the trained operator, not the test draw. Training minimises the top-20%-coefficient MSE (the headline elsewhere is top-10%); test PSNR is scored at four keep ratios ρ .

The random initialisations are genuinely different

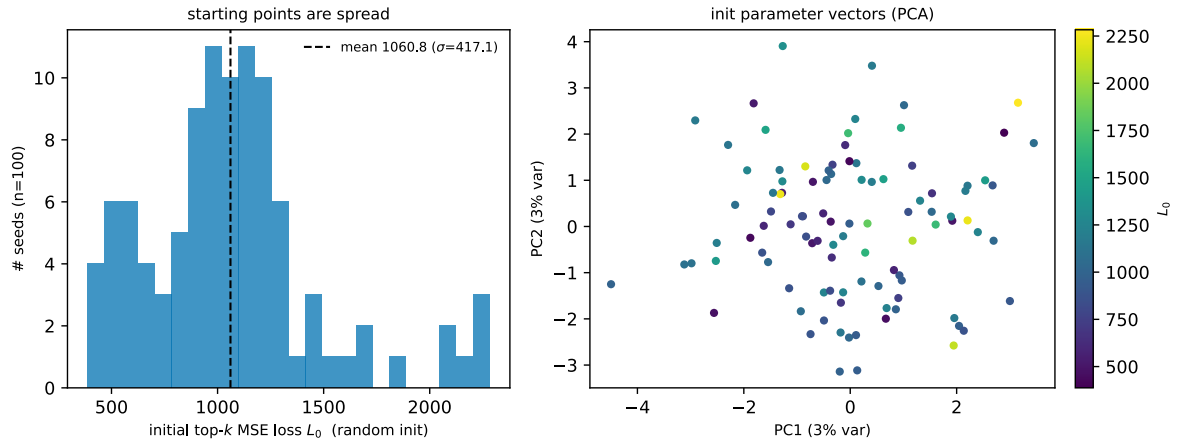


Figure 1: **Left:** the untrained random operator’s initial top- k MSE loss L_0 on a common fixed batch, per seed — it spans 387.3–2283.1 (mean 1060.8, $\sigma = 417.1$), so the seeds start from genuinely different points. **Right:** a 2-D PCA of the 100 init parameter vectors; the top two components explain only 2.8% + 2.6% of the variance, i.e. the inits are spread across many near-orthogonal directions rather than clustered.

Endpoint variance

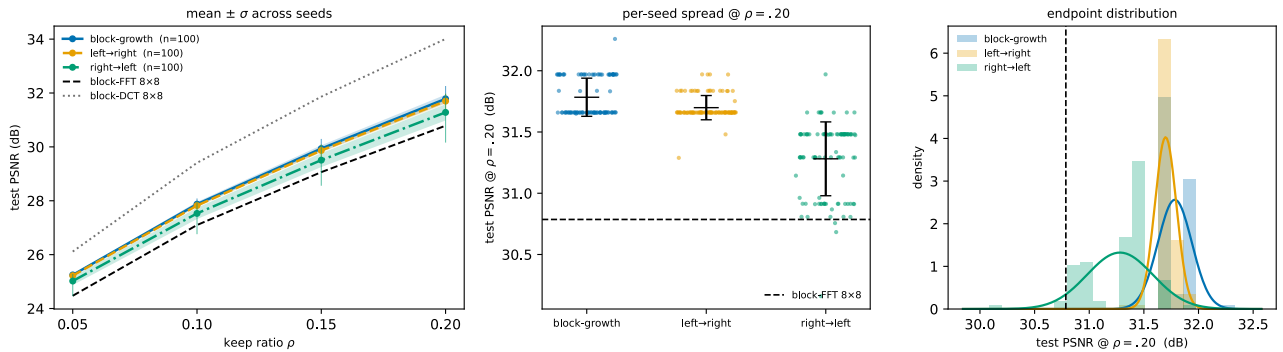


Figure 2: Per-ordering test PSNR across 100 seeds. **Left:** mean $\pm \sigma$ band (+ min–max whiskers) vs ρ , against the block-FFT 8×8 baseline (dashed) and block-DCT 8×8 (dotted). **Middle:** per-seed scatter at $\rho = \{.20\}$. **Right:** the endpoint distribution (histogram + fitted normal).

ordering	$\rho = \{.05\}$	$\rho = \{.10\}$	$\rho = \{.15\}$	$\rho = \{.20\}$	min@.20
block-growth	25.25$\pm$0.03	27.87$\pm$0.08	29.94$\pm$0.12	31.78$\pm$0.16	31.65

ordering	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$	min@.20
left→right	25.22±0.04	27.82±0.05	29.87±0.07	31.7±0.1	31.29
right→left	25.02±0.15	27.53±0.21	29.51±0.26	31.28±0.3	30.16
block-FFT 8×8	24.47	27.1	29.06	30.79	—
block-DCT 8×8	26.11	29.41	31.86	34.01	—

mean $\pm\sigma$ test PSNR (dB), $n = 100$ random-init seeds per ordering; **min@.20** is the single worst seed at $\rho\{=\}.20$.

Reading

Despite starting from 100 genuinely different random inits (previous section), each ordering’s endpoint sits in a narrow band: at $\rho\{=\}.20$ the per-seed standard deviation is block-growth 0.16, left→right 0.1, right→left 0.3 dB. The **release order**, not the random draw, is the dominant factor: block-growth and left→right are tight and sit highest, while right→left is $\sim 2\times$ wider and lowest — the dynamics figure below explains why (rl starts from a far worse loss and its seeds stay spread until the final unfreeze stages).

Against the classical block-FFT 8×8 reference (30.79 dB @ $\rho\{=\}.20$): block-growth and left→right clear it for **every** seed (worst seed 31.65 / 31.29 dB; margins +0.86 / +0.5 dB). right→left clears it for 97 of 100 seeds; 3 low-tail seeds dip below by at most 0.63 dB. Overall 297 of 300 runs (99%) clear block-FFT. So the gate-unfreeze QFT is **essentially always** above block-FFT — unconditionally for bg/lr, and for 97% of the highest-variance rl ordering — while block-DCT 8×8 (34.01 dB) remains the strongest classical transform here, ahead of the QFT family.

The per-seed endpoints are **not** Gaussian: a Shapiro–Wilk test rejects normality for all three orderings ($p < 10^{-7}$), because seeds settle into a few discrete attractor basins of the very-flat top- k MSE valley rather than scattering smoothly. The fitted normal in the endpoint panel is a visual summary of location and spread, not a claim of Gaussianity.

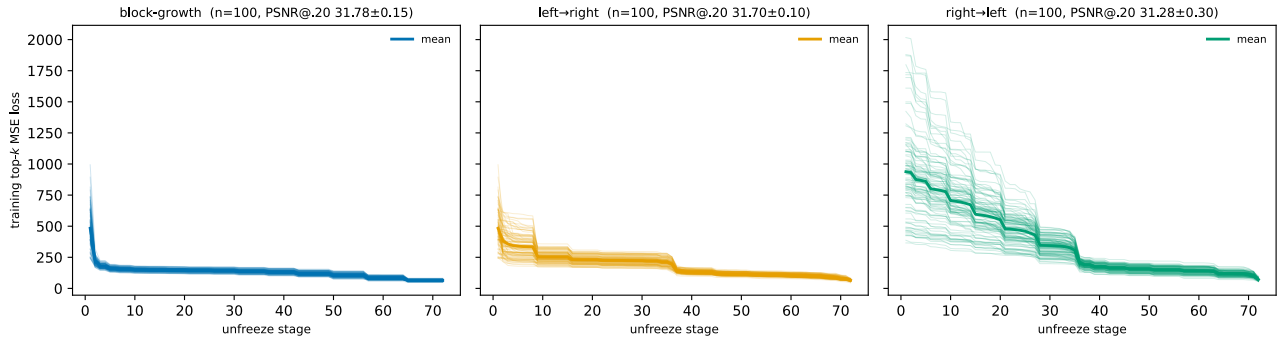


Figure 3: Per-seed training dynamics: every seed’s top- k MSE staircase (one per unfreeze stage), one panel per ordering. Different Haar starts descend along different paths: block-growth collapses fast and stays tight, whereas right→left starts far higher and its seeds remain spread until the final stages — visually accounting for the endpoint variances above.